

Rockchip RV Series Linux Audio Codec Developer Guide

ID: RK-KF-YF-934

Release Version: V1.5.0

Release Date: 2024-09-20

Security Level: ☐Top-Secret ☐Secret ☐Internal ☒Public

DISCLAIMER

THIS DOCUMENT IS PROVIDED "AS IS". ROCKCHIP ELECTRONICS CO., LTD. ("ROCKCHIP") DOES NOT PROVIDE ANY WARRANTY OF ANY KIND, EXPRESSED, IMPLIED OR OTHERWISE, WITH RESPECT TO THE ACCURACY, RELIABILITY, COMPLETENESS, MERCHANTABILITY, FITNESS FOR ANY PARTICULAR PURPOSE OR NON-INFRINGEMENT OF ANY REPRESENTATION, INFORMATION AND CONTENT IN THIS DOCUMENT. THIS DOCUMENT IS FOR REFERENCE ONLY. THIS DOCUMENT MAY BE UPDATED OR CHANGED WITHOUT ANY NOTICE AT ANY TIME DUE TO THE UPGRADES OF THE PRODUCT OR ANY OTHER REASONS.

Trademark Statement

"Rockchip", "瑞芯微", "瑞芯" shall be Rockchip's registered trademarks and owned by Rockchip. All the other trademarks or registered trademarks mentioned in this document shall be owned by their respective owners.

All rights reserved. ©2024 Rockchip Electronics Co., Ltd.

Beyond the scope of fair use, neither any entity nor individual shall extract, copy, or distribute this document in any form in whole or in part without the written approval of Rockchip.

Rockchip Electronics Co., Ltd.

No.18 Building, A District, No.89, software Boulevard Fuzhou, Fujian, PRC

Website: www.rock-chips.com

Customer service Tel: +86-4007-700-590

Customer service Fax: +86-591-83951833

Customer service e-Mail: fae@rock-chips.com

Preface

Overview

This document primarily introduces the features and functions of RV1106/RV1103 Audio Codec, as well as common attribute configurations of ACodec.

Product Version

Chipset	Kernel Version
RV1106/RV1103	5.10
RV1106B/RV1103B	5.10

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Revision History

Version	Author	Date	Change Description
V1.0.0	Xing Zheng	2022-05-12	Added the initial version of RV1106 Linux Audio Codec Developer Guide
V1.1.0	Xing Zheng	2022-08-31	Added commonly used playback and recording test commands of RV1106 Audio Codec
V1.2.0	Xing Zheng	2023-10-20	Fixed the <code>range</code> description error of ADC ALC node, added "DAC Control Manually" description
V1.3.0	Xing Zheng	2023-12-18	Fixed the <code>range</code> description error of ADC ALC node, and fixed the reference of the invalid document
V1.4.0	Xing Zheng	2024-07-07	Added the the introduction of RV1106B/RV1103B Audio Codec
V1.5.0	Xing Zheng	2024-09-20	Rename the document and unify the naming style

Contents

Rockchip RV Series Linux Audio Codec Developer Guide

1. Overview
 - 1.1 Function Introduction
2. ACodec Introduction
 - 2.1 ACodec Features
 - 2.2 ACodec Block Diagram
3. ACodec Software Configuration
 - 3.1 RV1106/RV1103 ACodec Kernel Configuration
 - 3.2 RV1106B/RV1103B ACodec Kernel Configuration
 - 3.3 RV1106/RV1103 ACodec User Level Configuration
 - 3.4 RV1106B/RV1103B ACodec User Level Configuration
4. ACodec Commonly Used Test Commands
 - 4.1 Commonly Used Playback Test Commands
 - 4.2 Commonly Used Capture Test Commands

1. Overview

1.1 Function Introduction

The ACodec mentioned here is the audio IP module built into the SoCs of RV1106/RV1103/RV1106B/RV1103B. It can be used to connect an external analog microphone and transmit the external signal recording to the CPU through analog-to-digital conversion. The CPU can also convert the local PCM audio digital signal through digital-to-analog conversion and then play it out.

The ACodec IP contained in RV1106 and RV1103 is exactly the same, including 2 ADCs and 1 DAC. The RV1106B/RV1103B SoC was launched in 2024, and the internal ACodec IP is exactly the same. Compared with RV1106/RV1103, its ACodec has deleted one ADC and is 1 ADC and 1 DAC. When operating ACodec, you will usually encounter the scenario of retrieving channel configuration, which will be explained in detail later in the document.

Note: If there is no special explanation, the following description of ACodec also applies to RV1106/RV1103/RV1106B/RV1103B.

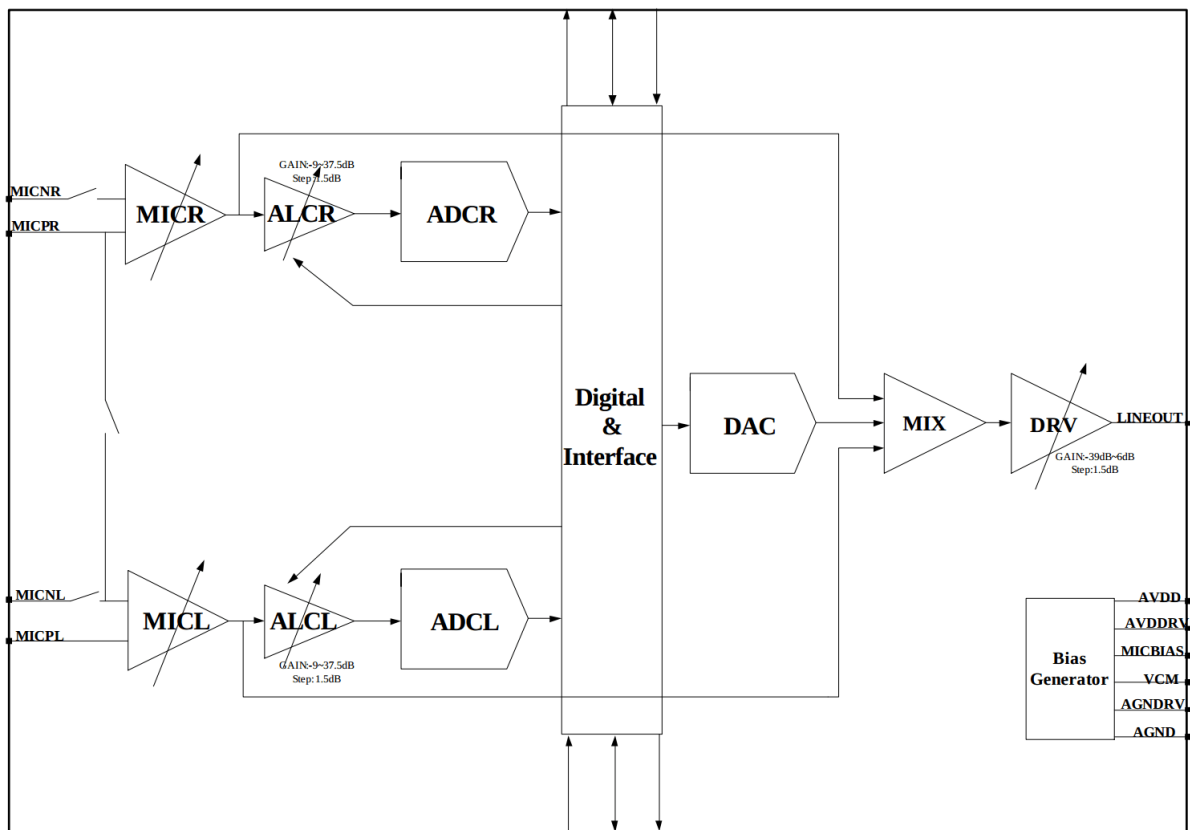
2. ACodec Introduction

2.1 ACodec Features

The basic features of RV1106/RV1103/RV1106B/RV1103B ACodec are as follows:

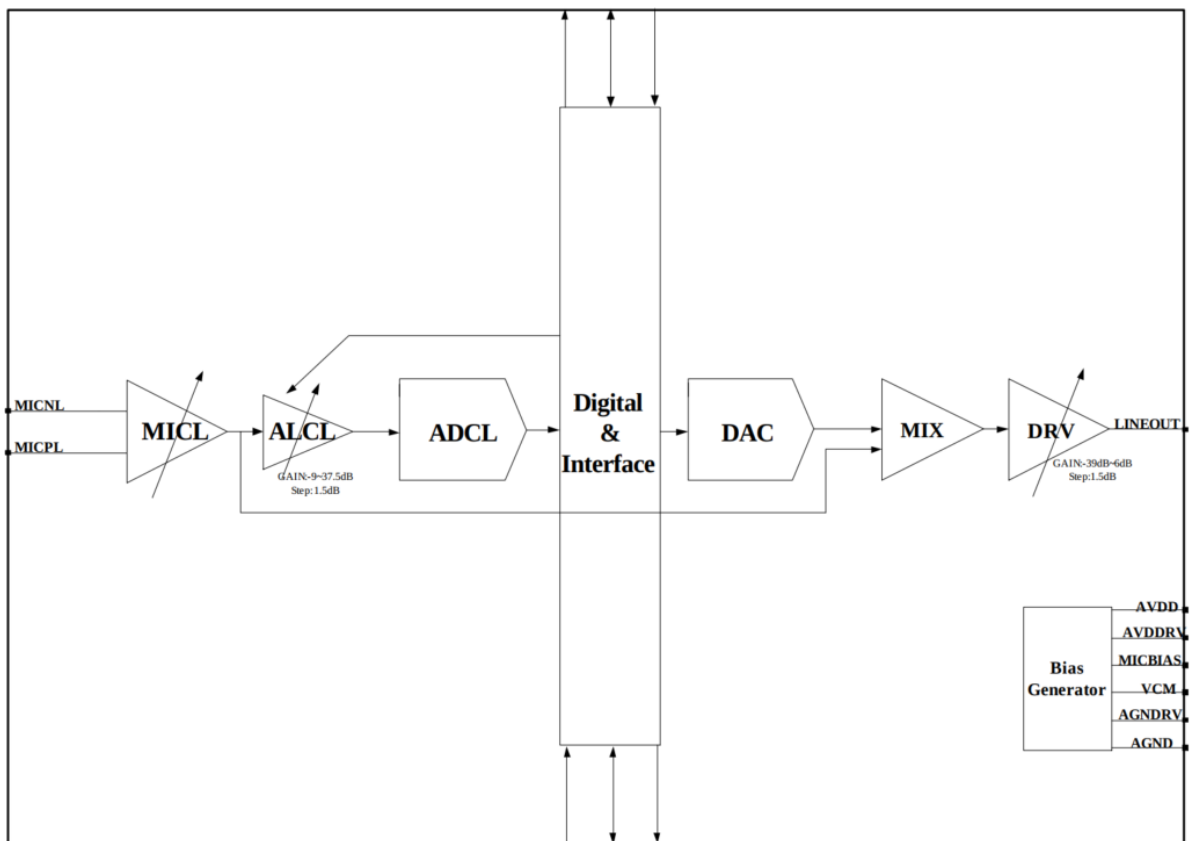
- 24 bits DAC with 93dB(A-weighted) SNR
- Support 600Ω line output
- Low power: 2.5mA for playback
- 24 bits ADC with 92dB(A-weighted) SNR
- Support differential and single-ended microphone or line input
- Low power: 5mA for stereo recording
- Automatic Level Control (ALC) for smooth audio recording
- Low power: less than 0.05mA for standby
- Programmable input and output analog gains
- Digital interpolation and decimation filter integrated
- Sampling rate of 8kHz/12kHz/16kHz/24kHz/32kHz/44.1KHz/48KHz/96KHz
- 1.8V supply for analog and 0.9V supply for digital

2.2 ACodec Block Diagram



RV1106/RV1103 ACodec Block Diagram

From the block diagram, it can be found that both RV1106 and RV1103 support 2 ADC inputs (for recording) and 1 DAC output (for playback). It's important to note that in differential mode, RV1106 can support a maximum of 2 microphones, while RV1103, in differential mode, only supports the connection of 1 microphone. In single-ended mode, both RV1106 and RV1103 can support a maximum of 2 microphones each.



RV1106B/RV1103B ACodec Block Diagram

The RV1106B/RV1103B ACodec removes one ADC in terms of function, and supports one ADC input (capture) and one DAC output (playback). However, the ADC input still maintains the two access modes of differential input or single-ended input.

Therefore, according to the SoC ID, the main differences of the following ACodec MIC access are:

SoC ID	Maximum number of MICs in single-ended mode	Maximum number of MICs in differential mode
RV1106	2	2
RV1103	2	1
RV1106B/RV1103B	1	1

For typical MIC connection methods, please refer to RK EVB drawings and hardware design reference guide.

3. ACodec Software Configuration

3.1 RV1106/RV1103 ACodec Kernel Configuration

Taking Rockchip kernel-5.10 SDK code as an example, you can refer to the following dtsi file:

```
arch/arm/boot/dts/rv1106-evb-v10.dtsi
```

Firstly, define an `acodec_sound` node, which describes some basic information of the sound card and references the DAI (Digital Audio Interface) `i2s0_8ch` on the controller side and the DAI `acodec` on the peripheral side:

```
acodec_sound: acodec-sound {
    compatible = "simple-audio-card";
    simple-audio-card,name = "rv1106-acodec";
    simple-audio-card,format = "i2s";
    simple-audio-card,mclk-fs = <256>;
    simple-audio-card,cpu {
        sound-dai = <&i2s0_8ch>;
    };
    simple-audio-card,codec {
        sound-dai = <&acodec>;
    };
};
```

Enable I2S0:

```
&i2s0_8ch {
    #sound-dai-cells = <0>;
    status = "okay";
};
```

Enable ACodec and define the GPIO pin that controls the external power amplifier:

```
&acodec {
    #sound-dai-cells = <0>;
    pa-ctl-gpios = <&gpio1 RK_PA1 GPIO_ACTIVE_HIGH>;
    status = "okay";
};
```

In the above example, the I2S0 inside RV1106/RV1103 can be bound to ACodec.

It should be noted that there is only one I2S controller inside RV1106/RV1103, which is I2S0. If the internal ACodec is enabled, I2S0 will be occupied at this time, and it is connected to ACodec through the SoC internal lead wires. Therefore, in this scenario, the pins connected to external IO by I2S0 will be unavailable.

3.2 RV1106B/RV1103B ACodec Kernel Configuration

The configuration of RV1106B/RV1103B ACodec can refer to

```
arch/arm/boot/dts/rv1103b-evb.dtsi
```

```
acodec_sound: acodec-sound {
    compatible = "simple-audio-card";
    simple-audio-card,name = "rv1103b-acodec";
    simple-audio-card,format = "i2s";
    simple-audio-card,mclk-fs = <256>;
    simple-audio-card,cpu {
        sound-dai = <&sai>;
    };
    simple-audio-card,codec {
        sound-dai = <&acodec>;
    };
};
```

It can be seen that the digital audio interface controller of RV1106B/RV1103B has been upgraded from I2S to SAI, which needs to be enabled:

```
&sai {
    status = "okay";
};
```

You also need to enable the acodec node:

```
&acodec {
    pa-ctl-gpios = <&gpio0 RK_PA3 GPIO_ACTIVE_HIGH>;
    status = "okay";
};
```

3.3 RV1106/RV1103 ACodec User Level Configuration

Usually, we can use the common and authoritative tinymix tool to obtain and set the control node information that the codec exposes to user layer. However, since tinymix tool was cropped in the SDK, rockit integrates the rk_mpi_amix_test tool that is equivalent to tinymix and can be used to list all contents information:

```
# rk_mpi_amix_test --list_contents
cmd parse result:
sound control id      : 0
control name          : (null)
control value         : (null)
list controls         : 0
list contents         : 1
Number of controls: 26
ctl    type    num    name                                     value
0      ENUM    1      I2STDM Digital Loopback Mode          ,
DisabledMode1Mode2Mode2 Swap
1      INT     1      ADC MIC Left Gain                     2 (range 0->3)
2      INT     1      ADC MIC Right Gain                    2 (range 0->3)
3      INT     1      ADC ALC Left Volume                   6 (range 0->31)
4      INT     1      ADC ALC Right Volume                  6 (range 0->31)
5      INT     1      ADC Digital Left Volume                195 (range 0-
>255)
6      INT     1      ADC Digital Right Volume              195 (range 0-
>255)
7      ENUM    1      ADC HPF Cut-off                       , OffOn
8      INT     1      ALC AGC Left Volume                   12 (range 0->31)
9      INT     1      ALC AGC Right Volume                  12 (range 0->31)
10     INT     1      ALC AGC Left Max Volume                7 (range 0->7)
11     INT     1      ALC AGC Right Max Volume               7 (range 0->7)
12     INT     1      ALC AGC Left Min Volume                0 (range 0->7)
13     INT     1      ALC AGC Right Min Volume               0 (range 0->7)
14     ENUM    1      ALC AGC Left Switch                   , OffOn
15     ENUM    1      ALC AGC Right Switch                  , OffOn
16     ENUM    1      AGC Left Approximate Sample Rate      ,
96KHz48KHz44.1KHz32KHz24KHz16KHz12KHz8KHz
17     ENUM    1      AGC Right Approximate Sample Rate     ,
96KHz48KHz44.1KHz32KHz24KHz16KHz12KHz8KHz
18     ENUM    1      ADC Mode                              ,
DiffadcLSingadcLDiffadcRSingadcRSingadcLRDiffadcLR
19     ENUM    1      ADC MICBIAS Voltage                   VREFx0_8VREFx0_825VREFx0_85VREFx0_875, VREFx0_9VREFx0_925VREFx0_95V
REFx0_975
20     ENUM    1      ADC Main MICBIAS                      Off, On
21     ENUM    1      ADC MIC Left Switch                   , WorkMute
22     ENUM    1      ADC MIC Right Switch                  , WorkMute
23     INT     1      DAC LINEOUT Volume                    26 (range 0->30)
24     INT     1      DAC HPMIX Volume                      1 (range 0->2)
25     ENUM    1      DAC Control Manually                  , NoneOffOn
```

There are lots of nodes, and it is going to introduce some commonly used node information here:

```
0      ENUM    1      I2STDM Digital Loopback Mode          ,
DisabledMode1Mode2Mode2 Swap
```


This node is actually the control node exposed to users by I2S0, but it is closely related to ACodec. It indicates whether the I2STDM controller is working in recovery mode. In the node information:

- Disabled: Default state, recovery mode is disabled
- Mode1: Suitable for 4ch usage scenarios. Channels 1-2 are data of MIC pickup, and channels 3-4 are recovery data.
- Mode2: Suitable for 2ch usage scenarios. The left channel is data of MIC pickup, and the right channel is the playback data of the right channel.
- Mode2 Swap: suitable for 2ch usage scenarios. The left channel is the recovery data of the left channel played, and the right channel is the MIC pickup data. Mode2 Swap and Mode2 channel order are opposite

Therefore, if Mode2 and Mode2 Swap are playing 2ch data, it is best to perform software MIX before sending data to ACodec DAC, so that the recovery reference signal is closer to the signal of the real echo path.

Remarks: the selected item displayed in tinymix style is added with a comma "," on the left. For example, the displayed ", Disabled" means that the "Disabled" state is currently selected, and so on.

1	INT	1	ADC MIC Left Gain	2 (range 0->3)
2	INT	1	ADC MIC Right Gain	2 (range 0->3)

ACodec ADC Boost Gain is the codec analog volume. The actual valid value range is 1 ~ 3:

- vol 0: Disabled and not recommended
- vol 1: 0dB
- vol 2: 20dB
- vol 3: 12dB

3	INT	1	ADC ALC Left Volume	6 (range 0->31)
4	INT	1	ADC ALC Right Volume	6 (range 0->31)

ACodec ADC ALC PGA Gain is the codec analog volume. The value range is 0~31:

- min: -9dB (vol: 0)
- max: +37.5dB (vol: 31)
- step: +1.5dB
- location: 0dB (vol: 6)

5	INT	1	ADC Digital Left Volume	195 (range 0->255)
6	INT	1	ADC Digital Right Volume	195 (range 0->255)

ACodec ADC Digital Gain is the codec digital volume. The value range is 0~255:

- min: -97.5dB (vol: 0)
- max: +30dB (vol: 255)
- step: +0.5dB
- location: 0dB (vol: 195)

18	ENUM	1	ADC Mode	, DiffadcLSingadcLDiffadcRSingadcRSingadcLRDiffadcLR
----	------	---	----------	---

"Diff" is the abbreviation of "Differential"; "Sing" is the abbreviation of "Single-end". Used to configure the ACodec ADC to work in differential or single-ended mode. The default is "Diff" differential mode. And in order to minimize power consumption to the greatest extent, the ADC only enables the L left channel. Therefore, "DiffadcL" is the default preference.

It should be noted that due to different chip orientation, RV1103 does not support two-channel pickup in differential mode, that is, it does not support the "DiffadcLR" option.

23	INT	1	DAC LINEOUT Volume	26 (range 0->30)
----	-----	---	--------------------	------------------

ACodec DAC Lineout Gain, the volume value range is 0~30:

- min: -39dB (vol: 0)
- max: +6dB (vol: 30)
- step: +1.5dB
- location: 0dB (vol: 26)

24	INT	1	DAC HPMIX Volume	1 (range 0->2)
----	-----	---	------------------	----------------

ACodec DAC HPMIX Gain, which is the preamp Gain of Lineout, is usually not set. The value range of the volume value is actually 1 and 2:

- vol 0: Disabled and not recommended
- vol 1: 0dB
- vol 2: 6dB

25	ENUM	1	DAC Control Manually	, NoneOffOn
----	------	---	----------------------	-------------

Developers usually use the media framework with the AO Stream is always on (always enabled the sound card). When stopping playback, they only need to quickly turn off and on the peripheral PA. Therefore, a "DAC Control Manually" node is added here:

- None: Default status, the media framework should be actually closed and AO Stream be enabled to control the sound card's off and on.
- Off: When the AO Stream is always on, you can just manually turn off the peripheral PA to make the speaker playback silent.
- On: When the AO Stream is always on and the previous state is Off, you can manually turn on the peripheral PA again to resume playback of the speaker.

The above are `codec control` node that are commonly used by developers during the development process. For other options, you can refer to the `control name` and options and configure and adjust them according to the specific project.

3.4 RV1106B/RV1103B ACodec User Level Configuration

We can also use `rk_mpi_amix_test` to list the control nodes related to RV1106B/RV1103B ACodec:

```
# rk_mpi_amix_test --list_contents
cmd parse result:
sound control id      : 0
control name          : (null)
control value         : (null)
list controls         : 0
```

```

list contents      : 1
Number of controls: 54
ctl    type    num    name                                     value
0      ENUM    1      SAI Transmit Start Mode Sel             ,
StandaloneChained
1      ENUM    1      SAI Receive Start Mode Sel              ,
StandaloneChained
2      ENUM    1      SAI Transmit SD0x Select                ,
AutoSD0x1SD0x2SD0x3SD0x4
3      ENUM    1      SAI Receive SDIx Select                ,
AutoSDIx1SDIx2SDIx3SDIx4
4      INT     1      SAI Receive Mono Slot Select          0 (range 0->128)
5      ENUM    1      SAI Receive Mono Switch              , DisableEnable
6      ENUM    1      SAI Transmit Mono Switch            , DisableEnable
7      ENUM    1      SAI SDI3 Loopback I2S LR Channel Sel  , L:MIC R:LPL:LP
R:MIC
8      ENUM    1      SAI SDI2 Loopback I2S LR Channel Sel  , L:MIC R:LPL:LP
R:MIC
9      ENUM    1      SAI SDI1 Loopback I2S LR Channel Sel  , L:MIC R:LPL:LP
R:MIC
10     ENUM    1      SAI SDI0 Loopback I2S LR Channel Sel  , L:MIC R:LPL:LP
R:MIC
11     ENUM    1      SAI SDI3 Loopback I2S LR Switch        , DisableEnable
12     ENUM    1      SAI SDI2 Loopback I2S LR Switch        , DisableEnable
13     ENUM    1      SAI SDI1 Loopback I2S LR Switch        , DisableEnable
14     ENUM    1      SAI SDI0 Loopback I2S LR Switch        , DisableEnable
15     ENUM    1      SAI SDI3 Loopback Src Select            , From SD00From
SD01From SD02From SD03
16     ENUM    1      SAI SDI2 Loopback Src Select            , From SD00From
SD01From SD02From SD03
17     ENUM    1      SAI SDI1 Loopback Src Select            , From SD00From
SD01From SD02From SD03
18     ENUM    1      SAI SDI0 Loopback Src Select            , From SD00From
SD01From SD02From SD03
19     ENUM    1      SAI SDI3 Loopback Switch              , DisableEnable
20     ENUM    1      SAI SDI2 Loopback Switch              , DisableEnable
21     ENUM    1      SAI SDI1 Loopback Switch              , DisableEnable
22     ENUM    1      SAI SDI0 Loopback Switch              , DisableEnable
23     ENUM    1      SAI Sync Out Switch                , From CRUFrom IO
24     ENUM    1      SAI Sync In Switch                  , From IOFrom
Sync Port
25     ENUM    1      SAI Receive PATH3 Source Select          From SDI0From
SDI1From SDI2, From SDI3
26     ENUM    1      SAI Receive PATH2 Source Select          From SDI0From
SDI1, From SDI2From SDI3
27     ENUM    1      SAI Receive PATH1 Source Select          From SDI0, From
SDI1From SDI2From SDI3
28     ENUM    1      SAI Receive PATH0 Source Select          , From SDI0From
SDI1From SDI2From SDI3
29     ENUM    1      SAI Transmit SD03 Source Select          From PATH0From
PATH1From PATH2, From PATH3
30     ENUM    1      SAI Transmit SD02 Source Select          From PATH0From
PATH1, From PATH2From PATH3
31     ENUM    1      SAI Transmit SD01 Source Select          From PATH0, From
PATH1From PATH2From PATH3
32     ENUM    1      SAI Transmit SD00 Source Select          , From PATH0From
PATH1From PATH2From PATH3
33     BOOL    1      SAI Clk Auto Switch                Off

```

34	INT	1	SAI PCM Read Wait Time MS	0 (range 0->10000)
35	INT	1	SAI PCM Write Wait Time MS	0 (range 0->10000)
36	INT	1	ADC MIC Left Gain	2 (range 0->3)
37	INT	1	ADC ALC Left Volume	6 (range 0->31)
38	INT	1	ADC Digital Left Volume	195 (range 0->255)
39	ENUM	1	ADC HPF Cut-off	, OffOn
40	INT	1	ALC AGC Left Volume	12 (range 0->31)
41	INT	1	ALC AGC Left Max Volume	7 (range 0->7)
42	INT	1	ALC AGC Left Min Volume	0 (range 0->7)
43	ENUM	1	ALC AGC Left Switch	, OffOn
44	ENUM	1	AGC Left Approximate Sample Rate	, 96KHz48KHz44.1KHz32KHz24KHz16KHz12KHz8KHz
45	ENUM	1	ADC Mode	, DiffadcLSingadcLDiffadcRSingadcRSingadcLRDiffadcLR
46	ENUM	1	ADC MICBIAS Voltage	VREFx0_8VREFx0_825VREFx0_85VREFx0_875, VREFx0_9VREFx0_925VREFx0_95VREFx0_975
47	ENUM	1	ADC Main MICBIAS	Off, On
48	ENUM	1	ADC MIC Left Switch	, WorkMute
49	INT	1	DAC LINEOUT Volume	26 (range 0->30)
50	INT	1	DAC HPMIX Volume	1 (range 0->2)
51	ENUM	1	DAC Control Manually	, NoneOffOn
52	ENUM	1	DAC VCM Manually	, NoneOffOn
53	ENUM	1	Audio Mux Select	, Path0Path1Path2Path3Path4Path5

As you can see, because the RV1106B/RV1103B ACodec has only one ADC input, the ADC "Right" part is removed compared to the RV1106/RV1103, leaving only the "Left" part, and the functional configuration is exactly the same as that of the RV1106/RV1103. A new "Audio Mux Select" function is added for the SAI controller to connect the internal ACodec and external IO channel switching function:

53	ENUM	1	Audio Mux Select	, Path0Path1Path2Path3Path4Path5
----	------	---	------------------	----------------------------------

There are 6 Path control states for channel switching:

- Path0: Play: Output only from ACodec Lineout; Record: Input only from ACodec Mic (default)
- Path1: Play: Output only from ACodec Lineout; Record: Input only from SAI SDI IO
- Path2: Play: Output from ACodec Lineout and SAI SDO IO at the same time; Record: Input only from ACodec Mic
- Path3: Play: Output from ACodec Lineout and SAI SDO IO at the same time; Record: Input only from SAI SDI IO
- Path4: Play: Output only from SAI SDO IO; Record: Input only from ACodec Mic
- Path5: Play: Output only from SAI SDO IO; Record: Input only from SAI SDI IO (for testing only, usually not used)

In addition, due to the upgrade of the controller SAI, the original RV1106/RV1103 The I2S part is quite different from the current SAI, but we mainly focus on the configuration related to the RV1106B/RV1103B digital loopback. Because the RV1103B/RV1106B ACodec has only one ADC input, the Data Lane related to the SAI input is only the SDI0 part:

10	ENUM	1	SAI SDI0 Loopback I2S LR Channel Sel	, L:MIC R:LPL:LP R:MIC
----	------	---	--------------------------------------	---------------------------

- L:MIC R:LP: SAI SDI0 Lane left and right channel selection for loopback mode: left channel is MIC, right channel is loopback
- L:LP R:MIC: SAI SDI0 Lane left and right channel selection for loopback mode: right channel is MIC, left channel is loopback

14	ENUM	1	SAI SDI0 Loopback I2S LR Switch	, DisableEnable
----	------	---	---------------------------------	-----------------

SAI SDI0 Lane left and right channel selection mode switch of the acquisition mode is disabled by default and remains in data pass-through mode. Enable it when digital acquisition is required.

18	ENUM	1	SAI SDI0 Loopback Src Select	, From SD00From SD01From SD02From SD03
----	------	---	------------------------------	---

Indicates the selection of the reference output data Lane after SAI SDI0 starts digital acquisition. Since RV1106B/RV1103B ACodec only supports mono recording and mono playback, only SDO0 is supported as a reference here. Keep the default state.

22	ENUM	1	SAI SDI0 Loopback Switch	, DisableEnable
----	------	---	--------------------------	-----------------

Digital retrieving switch of SAI SDI0 Lane. It is off by default and can be turned on when retrieving is needed.

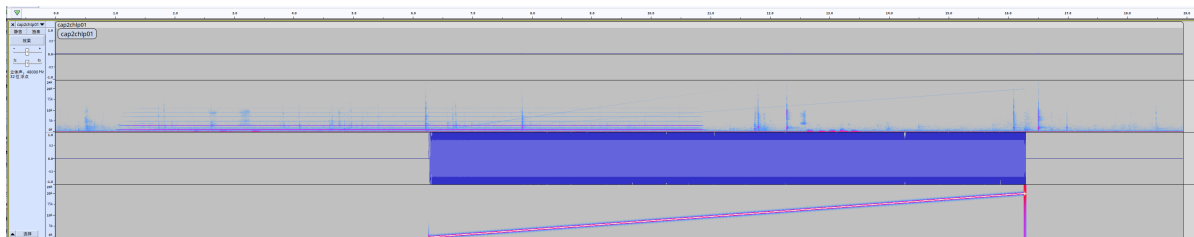
In summary, if you want to fully configure single MIC single retrieving, you can configure it according to the following command:

```
rk_mpi_amix_test --control "SAI SDI0 Loopback I2S LR Channel Sel" --value "L:MIC  
R:LP"  
rk_mpi_amix_test --control "SAI SDI0 Loopback Src Select" --value "From SD00"  
rk_mpi_amix_test --control "SAI SDI0 Loopback I2S LR Switch" --value "Enable"  
rk_mpi_amix_test --control "SAI SDI0 Loopback Switch" --value "Enable"
```

Since the "SAI SDI0 Loopback I2S LR Channel Sel" and "SAI SDI0 Loopback Src Select" registers maintain the required state after power-on, the configuration command for enabling digital acquisition can usually be omitted as follows:

```
rk_mpi_amix_test --control "SAI SDI0 Loopback I2S LR Switch" --value "Enable"  
rk_mpi_amix_test --control "SAI SDI0 Loopback Switch" --value "Enable"
```

RV1106B/RV1103B 2-channel capture example, the left channel is MIC, and the right channel is loopback:



4. ACodec Commonly Used Test Commands

Since RV1106/RV1103/RV1106B/RV1103B operating environment is usually based on the rockit framework, the following are combination of basic examples and rockit related commands.

4.1 Commonly Used Playback Test Commands

The following uses a PCM with a sampling rate of 16kHz/2ch/16bit as the playback sound source. The rk_mpi_ao_test command handles 16bit depth by default. in which, the parameters "device_rate=16000" and "input_rate=16000" are the same, indicating that the re-sampling function does not need to be enabled; the parameter "input_ch=2" needs to correspond to the original sound source 2ch:

```
rk_mpi_ao_test -i /tmp/sine_16000_500_1000_2ch.pcm --sound_card_name=hw:0,0 --  
device_ch=2 --device_rate=16000 --input_rate=16000 --input_ch=2
```

For more command parameter usage, please refer to the "MPI" related documents in the SDK Rockit project directory and the relevant test cases in the mpi example directory.

4.2 Commonly Used Capture Test Commands

The following uses PCM format recording with a sampling rate of 16kHz/2ch/16bit. The rk_mpi_ai_test command also processes 16 bit depth by default. in which, the parameters "device_rate=16000" and "input_rate=16000" are the same, indicating that the re-sampling function does not need to be enabled; the parameters "device_ch=2" and "input_ch=2" are the same, indicating that channel processing is not required. The data generated by recording is the cap_out.pcm file in the /tmp directory.

```
rk_mpi_ai_test --sound_card_name=hw:0,0 --device_rate=16000 --device_ch=2 --  
out_rate=16000 --out_ch=2 --output=/tmp
```

For more command parameter usage, please refer to the "AO/AI debugging steps" chapter in "Rockchip_Developer_Guide_MPI_FAQ.pdf".